pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Multiple Regression and Dummy Variables

WEEK 1 · DAY 2

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From one slope to a step for every rating

57.7 bp

test MAE, yesterday's line on rating score

35.5 bp

test MAE, rating dummies and duration, today

Today

Two features, then a rating as columns: one per level, one level left out and named.

821 bonds of invented issuers. Their spreads come from the course's own script: rating, maturity, an issuer effect, noise. Sector plays no part.

Every rating coefficient today is a difference from a reference level. Name it, or the number means nothing.

How multiple regression works

The model	Intercept plus one coefficient times each feature
The rule	Least squares: the smallest sum of squared misses on the training rows, all coefficients together
Held fixed	Each coefficient is its feature's effect with the other features held fixed
You control	Which features go in
What comes out	A coefficient in bp per unit of each feature, an intercept, and fitted values in bp

More than one feature: score and duration

```
LinearRegression().fit(train[["score", "duration"]], y_train)
```

Feature	Coefficient	Read as, other feature fixed
score	39.1432 bp	per notch of rating score
duration	0.8690 bp	per year of duration

	Training R2	Adjusted
score	0.5227	0.5220
+ duration	0.5230	0.5214

Test MAE 57.7 to 57.5 bp. Training R-squared always rises with a column; adjusted R-squared charges for it.

Holding the others fixed, in this data: a statement about the fitted spread, not about cause.

How dummy variables work

The problem	A rating is a category; a score forces equal steps between notches
Indicator columns	One column per level: 1.0 for a bond with that rating, 0.0 for every other bond
Reference level	One level is left out; the intercept stands for it
Each coefficient	The difference in bp from a bond of the reference level, other features held fixed
You control	The reference level: coefficients change, fitted spreads do not

A category as numbers: dummy variables

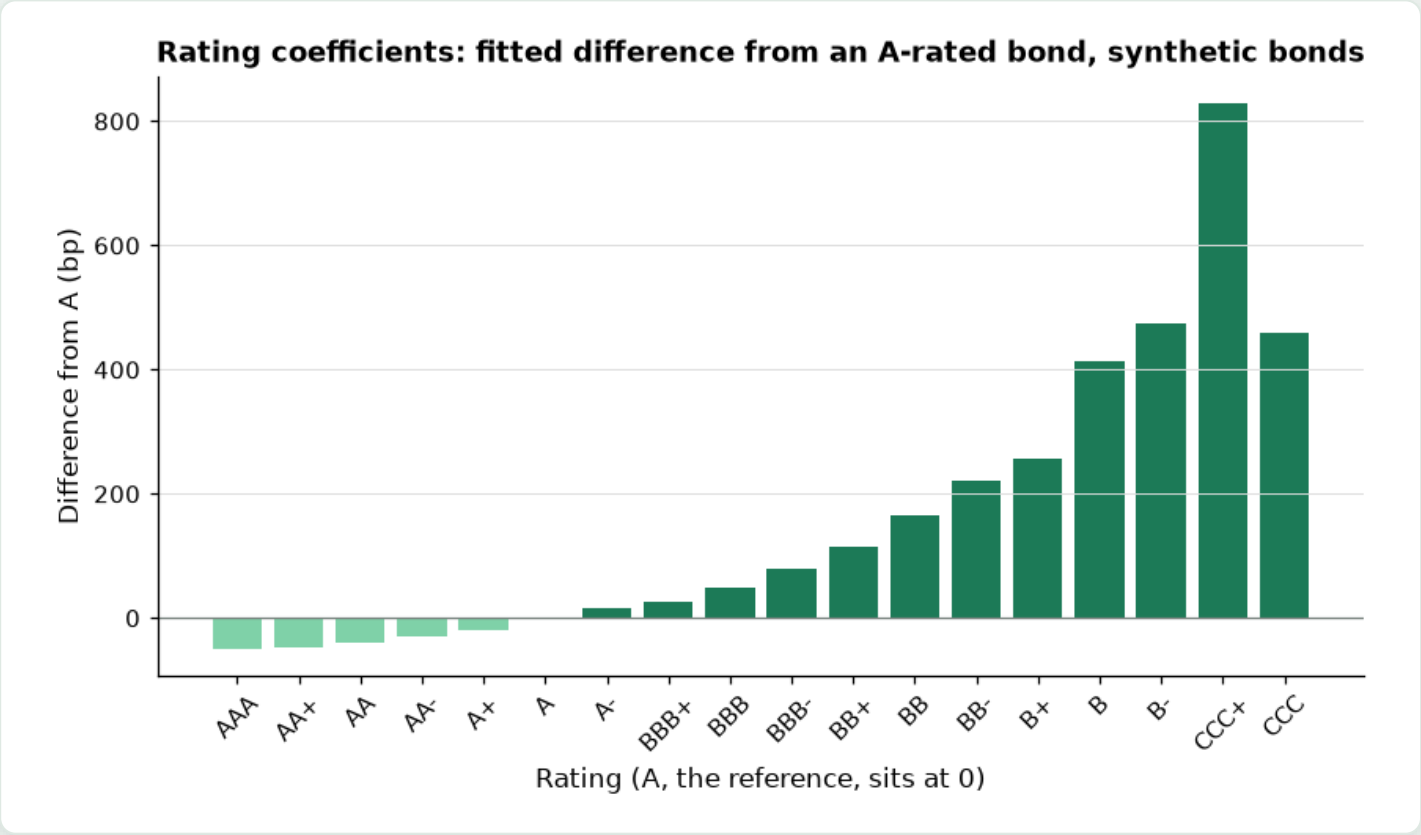
Bond	rating	rating_AAA	rating_BBB
1	AAA	1.0	0.0
2	A	0.0	0.0
3	BBB	0.0	1.0
4	A	0.0	0.0

The reference level: A

No column of its own: all zeros. The intercept stands for it, and every rating coefficient is a difference from an A-rated bond.

Which level is the reference is a choice. The fitted spreads do not depend on it; the coefficients do.

rating_dummies, with reference A



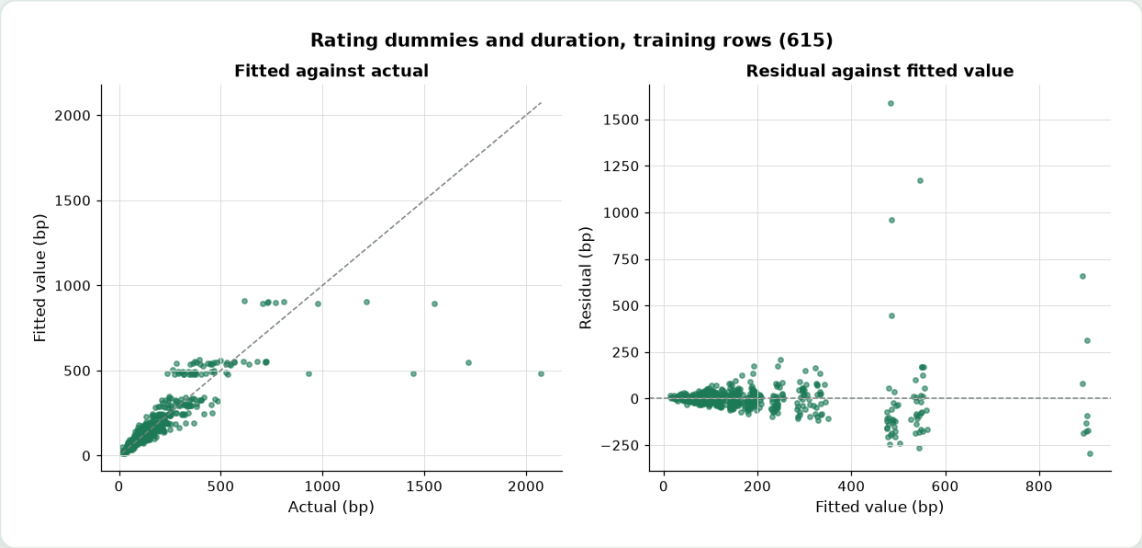

```
rating_dummies(r, levels, "A")
```

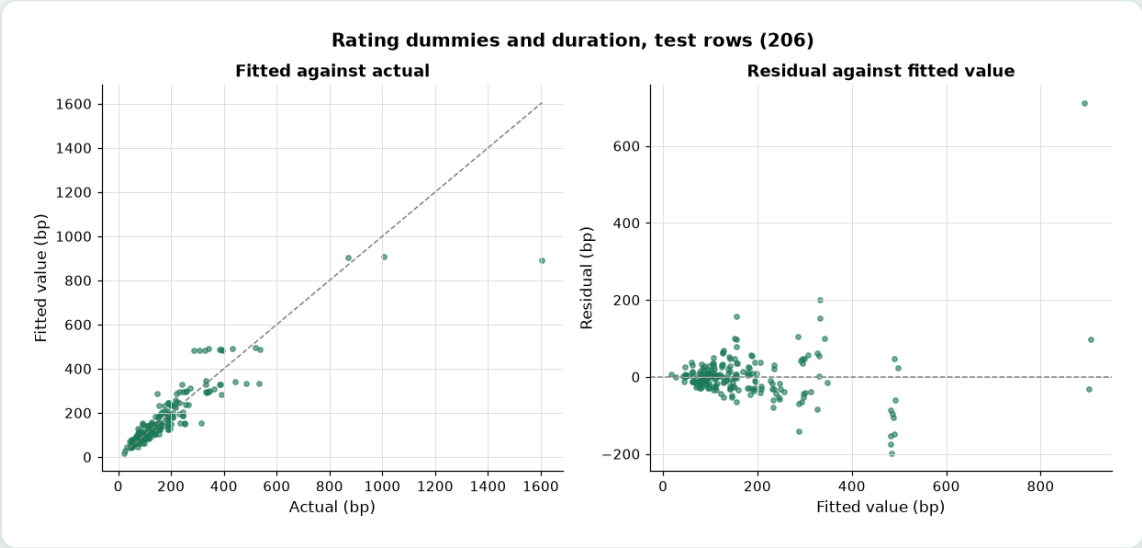
Rating	From A (bp)	Bonds
AAA	-49.0	11
BBB	48.0	42
BB	164.4	35
CCC+	828.7	9
CCC	458.1	3

Test MAE 35.5 bp, against 57.5 bp with the score.

A fitted spread, a difference from A, a residual. Never a verdict on a bond.

Fitted against actual: training and test rows





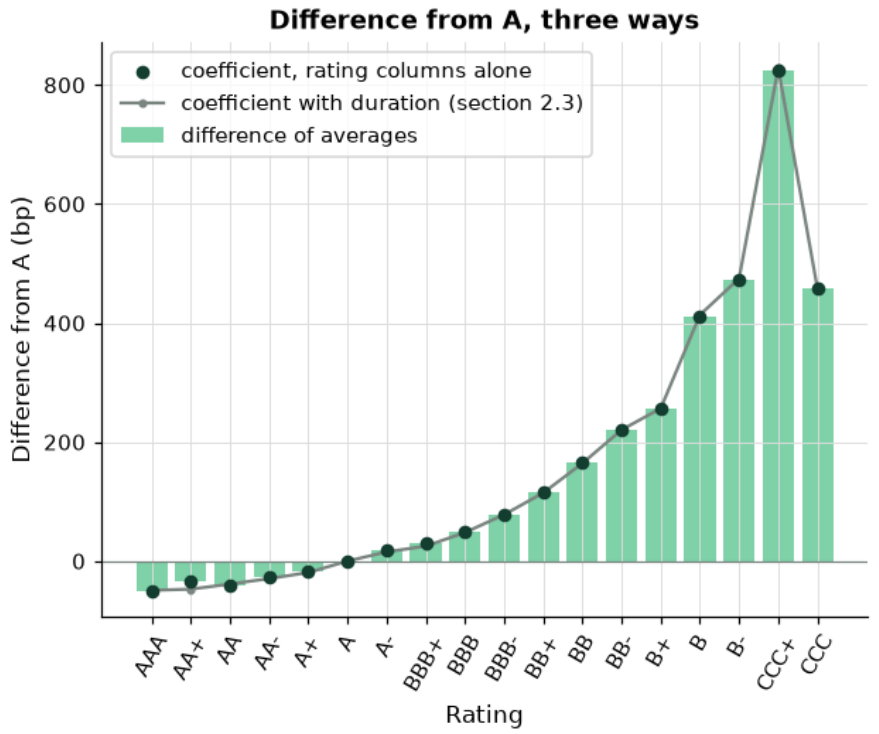
Rows	RMSE (bp)	MAE (bp)	MAPE (%)	R-squared
training	110.2	42.7	21.2	0.6727
test	69.1	35.5	19.8	0.8179

The big misses sit in the training rows, far above the zero line, and grow with the fitted spread.

One split put the hard bonds on the training side: the test row looks better than the fit.

A dummy coefficient is a difference of averages





Rating columns alone: the intercept is A's average (74.6 bp) and each coefficient a difference of averages (BBB 49.0 bp).

get_dummies(drop_first=True) drops A, not AAA

```
pd.get_dummies(bench["rating"], prefix="rating", drop_first=True)
levels sorted as text: ['A', 'A+', 'A-', 'AA', 'AA+'] ...
```

All bonds

Leaves out A: the course's reference, by luck.

High yield bonds

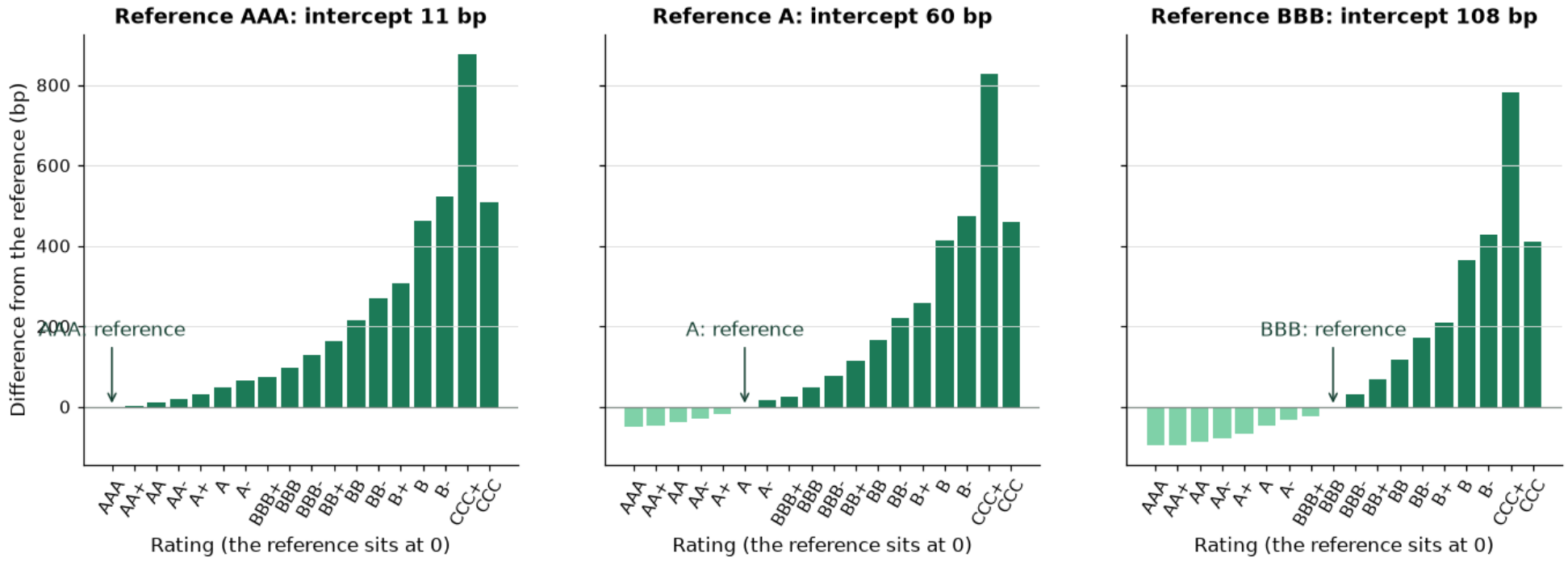
Leaves out B. Nothing in the code says so.

Train and test

17 columns against 14: no test bond is AA+, B-, CCC.

rating_dummies: levels from ratings.csv, the reference named, the same columns on every set of rows.

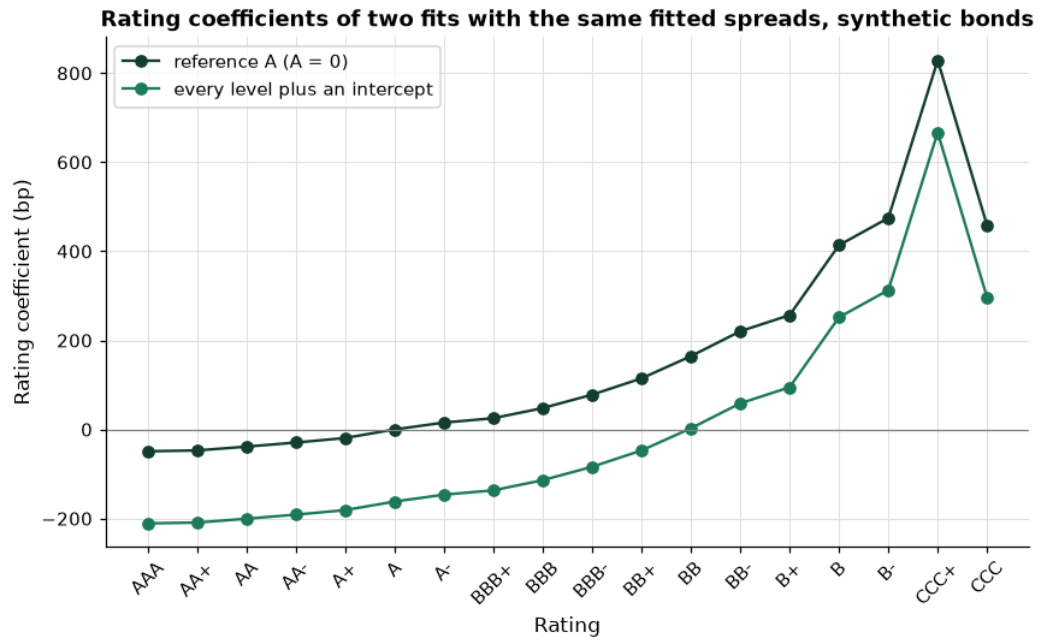
The reference changes the coefficients only



The fitted spread of every training bond is identical under the three references: a reporting choice, not a modelling one.

The dummy trap: every level plus an intercept

With a column for all 18 levels, the rating columns add up to 1.0 on every row: the intercept's column, twice.



The same fitted spreads

20 design columns, 19 independent.

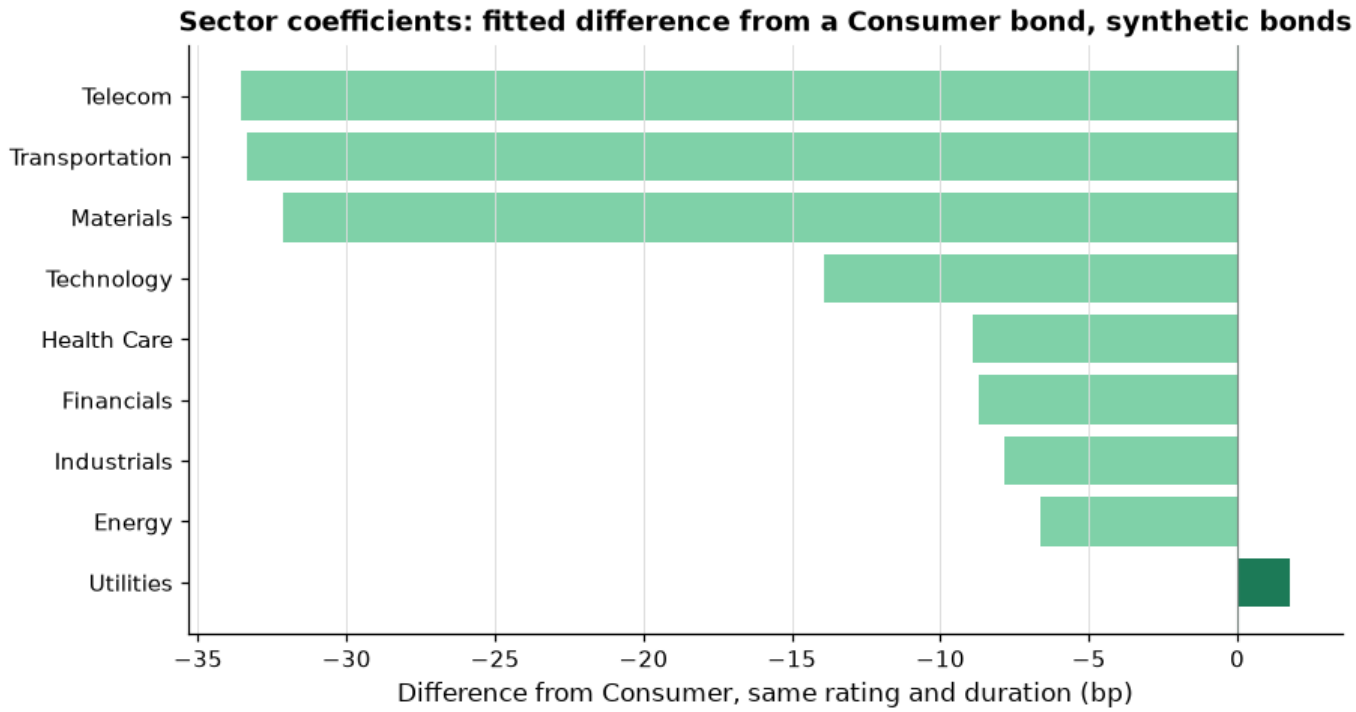
rating_A: -161.9 bp, a number with no reference.

rating_BB minus rating_A: 164.4 bp, as the reference-A model.

Excel's LINEST: no error; rating_CCC set to 0, so it chose CCC as the reference.

Two parallel lines: only differences between rating coefficients mean anything.

Sector dummies: a feature the generator never used



The true value: zero

The generator gives sector no part. Fitted: -33.5 to 1.8 bp.

On the test rows

Test MAE 36.3 bp with them, 35.5 bp without.

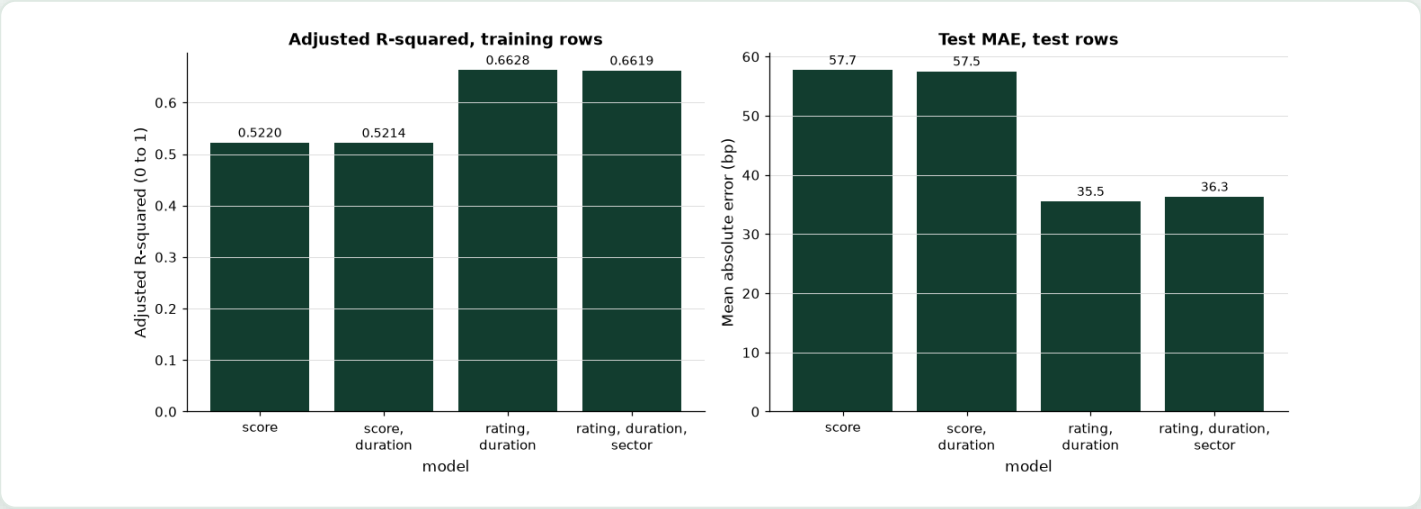
Nine columns fitted to noise in the training rows carry it to the test rows. Day 3 asks if they differ from zero.

How adjusted R-squared works

The problem	R-squared never falls when a feature is added, even a useless one
The charge	The unexplained share is scaled up by (rows - 1) over (rows - features - 1), then taken from 1
The effect	A feature must cut the misses by more than chance, or the number goes down
You control	Nothing: it needs R-squared, the rows, and the feature count
What comes out	A number at or below R-squared; read it beside the test MAE

Adjusted R-squared: a charge for each feature

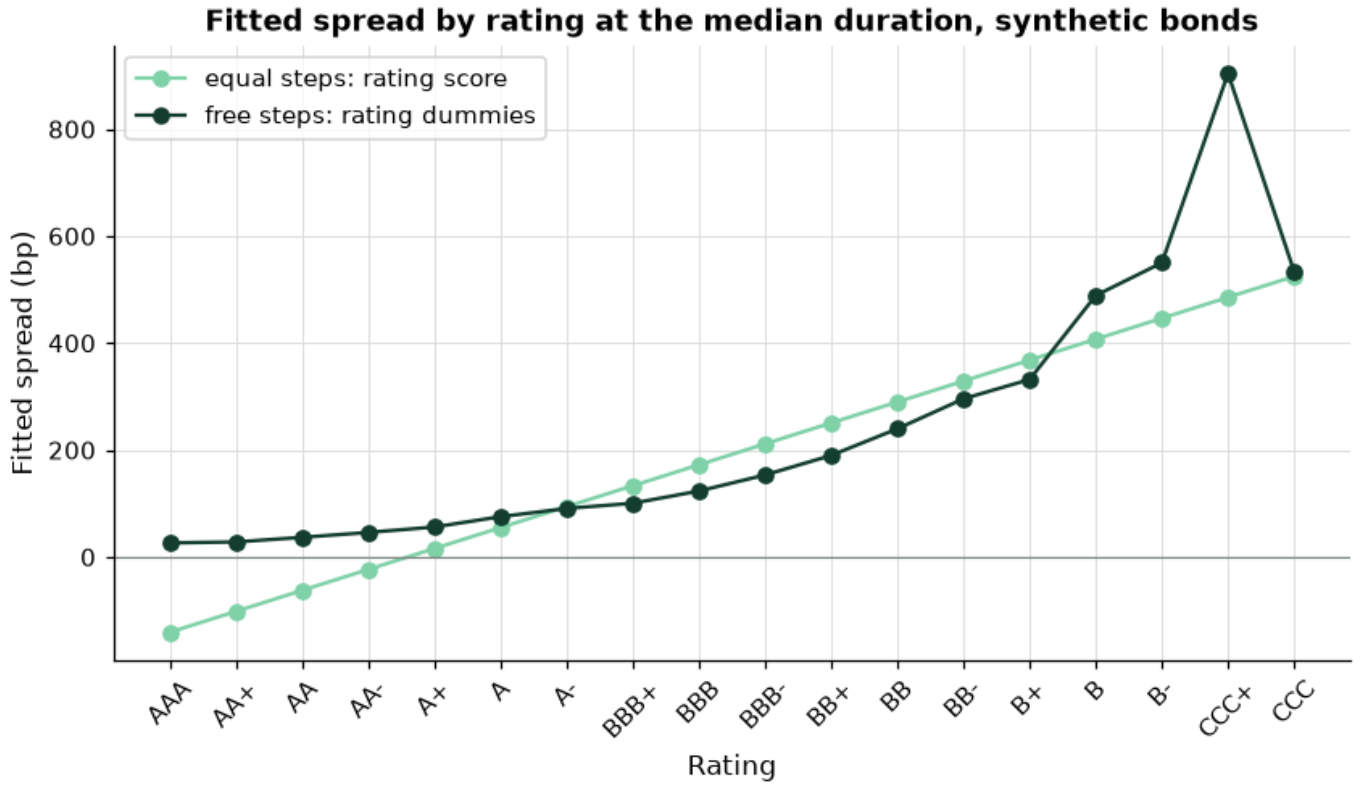
Model	Features	Training R2	Adjusted R2	Test MAE (bp)
score	1	0.5227	0.5220	57.7
score and duration	2	0.5230	0.5214	57.5
rating dummies and duration	18	0.6727	0.6628	35.5
rating, duration, and sector dummies	27	0.6768	0.6619	36.3




```
adj R2 = 1 - (1 - R2)
  x (n - 1) / (n - k - 1)
```

It falls twice, and the test MAE agrees. Still an in-sample number: the course chooses on unseen rows.

Equal steps against free steps



Equal steps: one slope

One coefficient on all 615 bonds. Below zero at the top; test MAE 57.5 bp.

Free steps: 17 coefficients

Each on its own level's bonds; test MAE 35.5 bp.

Thin levels wander: CCC (3 bonds) sits below CCC+. At 5.88 years' duration.

Excel side by side

Excel side by side (tested in Excel)

`=--($D2="AAA")  (G2, filled down)`	1 for AAA, 0 otherwise
`=LINEST(A2:A616,B2:C616,TRUE,TRUE)`	0.8690 | 39.1432 | -185.0334
`=LINEST(A2:A616,G2:X616,TRUE,TRUE)`	R-squared 0.6727
`=1-(1-INDEX(LINEST(...),3,1))*(ROWS(y)-1)/(ROWS(y)-k-1)`	adjusted 0.6628
`=TREND(train!y, train!X, G2:X2)  (test, filled down)`	the fitted spread
`=AVERAGE(ABS(A2:A207-AI2:AI207))`	test MAE 35.5 bp

LINEST's first row runs backwards: duration, score, then the intercept. Read it from the right.

Excel has no dummy function: each column is a formula. TREND is predict written as a formula.

Save the day: the experiment log moves

This morning: yesterday's module and log committed. Tonight: today's code, then git diff results.csv.

```
 model,features,train_rows,test_rows,test_r2,test_mae_bp,test_rmse_bp
 oas_on_score_line,score,615,206,0.5529,57.6620,108.3312
+oas_on_score_duration,score+duration,615,206,0.5529,57.4884,108.3237
+oas_on_rating_duration,rating(ref A)+duration,615,206,0.8179,35.5395,69.1351
+oas_on_rating_duration_sector,rating(ref A)+duration+sector(ref Consumer),615,206,0.8149,36.3359,69.7107
```

One context line, three added lines: the test MAE column reads the day.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: get_dummies(drop_first=True) drops A, but the summary calls every coefficient "the spread over AAA".

Code that runs is not the same as code that is right.

What you can do now

- Each coefficient is read with the other features fixed
- A category becomes columns; one level is the named reference
- drop_first leaves out whatever sorts first
- The dummy trap runs without an error
- Adjusted R-squared charges; the test rows decide

Week 1, Day 3 Coefficients, p-values, and multicollinearity: statsmodels and sector's p-values